EMC First Midterm — Practice Questions

Course: Electromagnetic Compatibility, NTUST
Format: Closed-book midterm
All citations refer to: chap1.pdf – chap4.pdf

Chapter 1: Introduction to EMC

Conceptual Questions

Q1.1 Define electromagnetic compatibility (EMC). What are the three criteria a system must satisfy to be considered electromagnetically compatible? *(chap1.pdf, p.4)*

Q1.2 Describe the three-element model of EMC coupling. What are the three ways to prevent electromagnetic interference, and why is suppressing the source the most preferred approach? *(chap1.pdf, p.5–6)*

Q1.3 List the four basic EMC subproblems and give a brief description of each. *(chap1.pdf, p.7)*

Q1.4 A PCB trace has a physical length of 5 cm. At 1 GHz, would you analyze it using lumped-circuit theory or distributed (transmission-line) theory? Justify your answer using the concept of electrical dimension. Assume the substrate is air ($\epsilon_r = 1$). *(chap1.pdf, p.21)*

Q1.5 Explain why, for an EMC engineer, a signal's **electrical dimension** is more important than its physical dimension. *(chap1.pdf, p.15, 21)*

Short Calculations — Chapter 1

Q1.6 A signal propagates along a PCB trace of length $\mathscr{L} = 20$ cm through a medium with $\epsilon_r = 4.0$, $\mu_r = 1$.

(a) Find the propagation velocity v.

(b) Find the time delay T_D.

(c) At $f = 750$ MHz, find the wavelength λ in the medium.

(d) Compute the electrical dimension $k = \mathscr{L}/\lambda$.

(chap1.pdf, p.16, 18, 20, 21)

Q1.7 Convert the following:

(a) A voltage ratio of 100 to dB.

(b) A power ratio of 0.001 to dB.

(c) -14 dBm to dBμV (50 Ω system).

(d) 87 dBμV to dBm (50 Ω system).

(chap1.pdf, p.25–27, 30)

Q1.8 A cable has an attenuation constant $\alpha = 0.08$ Np/m and length $\mathscr{L} = 5$ m. Find the cable loss in dB. *(chap1.pdf, p.30)*

Q1.9 Three amplifier stages are cascaded with gains 20 dB, -6 dB, and 14 dB. What is the total gain in dB? What is the output power in dBm if the input is -10 dBm? *(chap1.pdf, p.28)*

True/False — Chapter 1

Q1.10 In a denser dielectric (higher ϵ_r), the wavelength is longer. **T / F** *(chap1.pdf, p.20–21)*

Q1.11 Doubling the voltage gain doubles the gain in dB. **T / F** *(chap1.pdf, p.26)*

Q1.12 The time delay $T_D = \mathscr{L}/v$ depends on the signal frequency. **T / F** *(chap1.pdf, p.16)*

Q1.13 Suppressing the emission at its source is the most cost-effective EMC solution. **T / F** *(chap1.pdf, p.6)*

Multiple Choice — Chapter 1

Q1.14 The electrical dimension $k = \mathscr{L}/\lambda$ of a 10 cm conductor at 3 GHz in air is approximately:
(A) 0.01 (B) 0.1 (C) 1.0 (D) 10

(chap1.pdf, p.21)

Q1.15 When $R_{\text{in}} = R_L$, the power gain in dB equals:
(A) $10\log_{10}(V_{\text{out}}/V_{\text{in}})$
(B) $20\log_{10}(V_{\text{out}}/V_{\text{in}})$
(C) $10\log_{10}(V_{\text{out}}^2/V_{\text{in}}^2)$
(D) Both (B) and (C)

(chap1.pdf, p.26)

Chapter 2: EMC Requirements

Conceptual Questions

Q2.1 What is the difference between FCC Class A and Class B digital devices? Which has more stringent limits and why? *(chap2.pdf, p.6)*

Q2.2 Describe the two types of FCC emissions that digital devices must comply with. State the frequency ranges for each. *(chap2.pdf, p.7)*

Q2.3 What is a LISN (Line Impedance Stabilization Network)? Describe its two functions and why the 50 Ω impedance it presents is important. *(chap2.pdf, p.29, 32)*

Q2.4 Compare FCC and CISPR 22 radiated emission limits: what frequency range does each cover, and at what measurement distance? *(chap2.pdf, p.11, 14)*

Q2.5 MIL-STD-461E requirements can be tailored or waived by the procuring authority. Why is this different from FCC or CISPR 22 regulations? *(chap2.pdf, p.17)*

Short Calculations — Chapter 2

Q2.6 The FCC Class A radiated emission limit at 30–88 MHz is 39 dBμV/m measured at 10 m. Convert this to a 3 m measurement distance and compare it to the Class B limit at 3 m (40 dBμV/m). Which is more stringent at 3 m? *(chap2.pdf, p.11, 13)*

Q2.7 A product passes the FCC Class B radiated emission limit at 10 m with a margin of 5 dB. What margin does it have at 3 m (i.e., does passing at 10 m guarantee passing at 3 m)? *(chap2.pdf, p.13)*

Q2.8 A digital product generates harmonics at 74.9 MHz, 134.8 MHz, and 174.8 MHz. The FCC Class B radiated limit at 3 m for the frequency range 88–216 MHz is 43.5 dBμV/m. A measurement shows 38.5 dBμV/m at 158.8 MHz. By how many dB does it pass or fail? *(chap2.pdf, p.11, 33)*

True/False — Chapter 2

Q2.9 CISPR 22 conducted emission limits are the same as FCC Part 15 conducted emission limits. **T / F** *(chap2.pdf, p.15)*

Q2.10 The FCC Class B limit is measured at 3 m; the Class A limit is measured at 10 m. When normalized to the same distance, Class A is less stringent. **T / F** *(chap2.pdf, p.13)*

Q2.11 CISPR 22 radiated emissions cover frequencies up to 40 GHz. **T / F** *(chap2.pdf, p.14)*

Q2.12 MIL-STD-461E RS103 requires an electric field immunity of up to 200 V/m for aircraft. **T / F** *(chap2.pdf, p.20)*

Multiple Choice — Chapter 2

Q2.13 The frequency range for FCC conducted emission measurements is:
(A) 30 Hz – 30 MHz
(B) 150 kHz – 30 MHz
(C) 30 MHz – 1 GHz
(D) 10 kHz – 40 GHz

(chap2.pdf, p.7)

Q2.14 Moving a measurement antenna from 10 m to 3 m changes the measured field level by approximately:
(A) +3.3 dB (B) +10 dB (C) +10.5 dB (D) -10.5 dB

(chap2.pdf, p.13)

Chapter 3: Signal Spectra

Conceptual Questions

Q3.1 Explain why the **rise time** of a digital signal, rather than its clock frequency, dominates the high-frequency content of its spectrum. Use the spectral bound breakpoints to support your answer. *(chap3.pdf, p.3, 34)*

Q3.2 A trapezoidal clock signal has breakpoints $f_1 = 1/(\pi\tau)$ and $f_2 = 1/(\pi\tau_r)$. Sketch the spectral envelope on a log-log (Bode-style) plot, labeling the slopes in each region. *(chap3.pdf, p.34)*

Q3.3 For a 50% duty cycle clock signal, why are even harmonics theoretically zero? *(chap3.pdf, p.14, 31)*

Q3.4 Reducing the duty cycle of a clock signal (shorter pulse width) lowers the DC level but does not change the high-frequency content of the spectrum. Explain why. *(chap3.pdf, p.39)*

Q3.5 What is the difference between a peak detector and a quasi-peak detector in a spectrum analyzer? Why does the regulatory standard use quasi-peak? *(chap3.pdf, p.46)*

Q3.6 Describe the power spectral density (PSD) of a PCM-NRZ data signal. What does the delta function at DC represent physically? *(chap3.pdf, p.52)*

Short Calculations — Chapter 3

Q3.7 A trapezoidal clock waveform has: $A = 5$ V, $T = 20$ ns, $\tau = 8$ ns, $\tau_r = 2$ ns.

(a) Find f_0, f_1, f_2, and BW.

(b) Find the DC (average) level c_0.

(c) State the slope of the envelope for $f > f_2$.

(chap3.pdf, p.34, 37)

Q3.8 For a rectangular pulse train with $A = 3$ V, $\tau = 10$ ns, $T = 100$ ns:

(a) Find $|c_n|$ at $n = 1$ ($f = 10$ MHz).

(b) Find $|c_n|$ at $n = 5$ ($f = 50$ MHz).

(chap3.pdf, p.12)

Q3.9 A PCM-NRZ signal has bit period $T = 1$ ns and amplitude $X_0 = 3.3$ V.

(a) Find the bit rate (in Gbps).

(b) At what frequency is the first null in the sinc-squared component of the PSD?

(c) Write the PSD $G_x(f)$ with numerical values substituted.

(chap3.pdf, p.52)

True/False — Chapter 3

Q3.10 Increasing the rise time of a clock signal (making edges slower) reduces its high-frequency emissions. **T / F** *(chap3.pdf, p.35)*

Q3.11 For a PCM-NRZ signal with equal probability of 0 and 1, the PSD has a DC delta function. **T / F** *(chap3.pdf, p.52)*

Q3.12 The bandwidth of a digital signal is BW $= 1/T$ where T is the clock period. **T / F** *(chap3.pdf, p.37)*

Q3.13 For a 50% duty cycle clock, odd harmonics are zero and even harmonics are nonzero. **T / F** *(chap3.pdf, p.14)*

Multiple Choice — Chapter 3

Q3.14 A clock signal has rise time $\tau_r = 0.5$ ns. Its bandwidth is approximately:
(A) 0.5 GHz (B) 1 GHz (C) 2 GHz (D) 4 GHz

(chap3.pdf, p.37)

Q3.15 The slope of the trapezoidal spectral bound above the second breakpoint f_2 is:
(A) 0 dB/decade (B) -20 dB/decade (C) -40 dB/decade (D) -60 dB/decade

(chap3.pdf, p.34)

Q3.16 If the pulse width τ is doubled (duty cycle doubled) while keeping τ_r constant, the second spectral breakpoint f_2:
(A) doubles (B) halves (C) stays the same (D) quadruples

(chap3.pdf, p.34)

Chapter 4: Transmission Lines

Conceptual Questions

Q4.1 Write the two Telegrapher's equations (first-order coupled form) for a lossless transmission line and identify the physical law (KVL or KCL) behind each. *(chap4.pdf, p.8–9)*

Q4.2 What does it mean for a transmission line to support a TEM wave? What restriction does this place on field components in the direction of propagation? *(chap4.pdf, p.11)*

Q4.3 Explain the physical meaning of the characteristic impedance $Z_C = \sqrt{l/c}$. Why is it important for signal integrity? *(chap4.pdf, p.28, 34)*

Q4.4 For each of the three special load cases (open circuit, short circuit, matched), state the reflection coefficient Γ_L and describe what happens to an incident pulse. *(chap4.pdf, p.35)*

Q4.5 What is the criterion for a transmission line to be negligible (treated as a simple wire)? Express it in terms of rise time and time delay. *(chap4.pdf, p.59)*

Q4.6 Compare series termination and parallel termination for eliminating reflections. What are the trade-offs? *(chap4.pdf, p.54–55)*

Q4.7 Explain what happens to conductor resistance as frequency increases, and why. How does skin depth relate to this? *(chap4.pdf, p.82–83)*

Q4.8 Define loss tangent $\tan\theta$ and explain how dielectric loss (conductance g) differs from skin-effect loss (resistance r) in terms of frequency dependence. *(chap4.pdf, p.85–87)*

Short Calculations — Chapter 4

Q4.9 A transmission line has $Z_C = 75\,\Omega$ and is driven by a source with $R_S = 150\,\Omega$ and $V_S = 6$ V (step). The load is $R_L = 25\,\Omega$.

(a) Find Γ_L and Γ_S.

(b) Find the initially launched voltage V^+.

(c) Find the load voltage immediately after the first arrival ($t = T_D^+$).

(chap4.pdf, p.35, 37)

Q4.10 A lossless 50 Ω line has $v = 2 \times 10^8$ m/s and $\mathscr{L} = 0.5$ m. At $f = 1$ GHz, find:

(a) The phase constant β.

(b) $\beta\mathscr{L}$ in radians.

(c) The input impedance for an **open-circuit** load ($Z_L \to \infty$).

(chap4.pdf, p.74, 76)

Q4.11 A 50 Ω transmission line is terminated with $Z_L = 100 + j50\,\Omega$. Find $|\Gamma_L|$ and VSWR. *(chap4.pdf, p.74, 77)*

Q4.12 Check the transmission-line negligibility criterion for:

(a) $\tau_r = 5$ ns, $\mathscr{L} = 10$ cm, $v = 2 \times 10^8$ m/s, $\epsilon_r' \approx 2.25$.

(b) $\tau_r = 0.5$ ns, same line.

(chap4.pdf, p.59)

Q4.13 A stripline PCB trace has $Z_C = 50\,\Omega$ and $\epsilon_r = 4.0$. A CMOS gate with output resistance $R_S = 20\,\Omega$ drives it. What series resistor value should be added to achieve series matching? *(chap4.pdf, p.54)*

Q4.14 Find the skin depth in copper ($\sigma = 5.8 \times 10^7$ S/m) at:

(a) $f = 1$ MHz

(b) $f = 1$ GHz

How does skin depth change between these frequencies?

(chap4.pdf, p.82)

Q4.15 A lossy transmission line has $r = 0.5\,\Omega$/m, $g = 10^{-4}$ S/m, $Z_C = 50\,\Omega$. Using the low-loss approximation, find α. Then find the cable loss for $\mathscr{L} = 3$ m. *(chap4.pdf, p.91, 89)*

True/False — Chapter 4

Q4.16 For an open-circuit load, the voltage reflection coefficient is $\Gamma_L = +1$. **T / F** *(chap4.pdf, p.35)*

Q4.17 The product lc depends on the geometry of the transmission line, not the surrounding medium. **T / F** *(chap4.pdf, p.12)*

Q4.18 A matched load ($R_L = Z_C$) eliminates reflections and produces a clean time-delayed replica of the source waveform at the load. **T / F** *(chap4.pdf, p.43)*

Q4.19 VSWR = 1 corresponds to a perfectly matched line. **T / F** *(chap4.pdf, p.77)*

Q4.20 Skin depth increases with frequency ($\delta \propto \sqrt{f}$). **T / F** *(chap4.pdf, p.82)*

Q4.21 In the low-loss approximation, the phase constant $\beta \approx \omega\sqrt{lc}$, which is linear in frequency — meaning no phase dispersion. **T / F** *(chap4.pdf, p.91)*

Multiple Choice — Chapter 4

Q4.22 A 50 Ω line is short-circuit terminated. The reflection coefficient and VSWR are:
(A) $\Gamma_L = +1$, VSWR = ∞
(B) $\Gamma_L = -1$, VSWR = ∞
(C) $\Gamma_L = 0$, VSWR = 1
(D) $\Gamma_L = -1$, VSWR = 1

(chap4.pdf, p.35, 77)

Q4.23 The telegrapher's equation $\partial V/\partial z = -l\,\partial I/\partial t$ is derived by applying:
(A) KCL at a node
(B) KVL around a loop
(C) Gauss's law
(D) Faraday's law in integral form

(chap4.pdf, p.8)

Q4.24 A signal with rise time $\tau_r = 2$ ns travels on a line with $T_D = 0.1$ ns. The line:
(A) Must be analyzed as a transmission line
(B) Can be treated as a lumped element ($\tau_r > 10T_D$)
(C) Requires Maxwell's equations
(D) None of the above

(chap4.pdf, p.59)

Q4.25 At high frequencies, the resistance per unit length r of a wire scales approximately as:
(A) Constant (independent of f)

(B) $\propto f$
(C) $\propto \sqrt{f}$
(D) $\propto f^2$

(chap4.pdf, p.83)

Answers

Chapter 1

A1.1 A system is EMC when it: (1) does not cause interference with others; (2) is not susceptible to interference from others; (3) does not interfere with itself. *(chap1.pdf, p.4)*

A1.2 Source $\rightarrow$ Path $\rightarrow$ Receptor. Three prevention methods: (1) suppress emission at source, (2) attenuate the coupling path, (3) make receptor less susceptible. Fixing the source is cheapest and most effective because it prevents the problem before it propagates. *(chap1.pdf, p.5–6)*

A1.3 (1) Radiated emissions: noisy component radiates outward. (2) Radiated susceptibility: external radiation couples in. (3) Conducted emissions: noise exits via cable. (4) Conducted susceptibility: noise enters via cable. *(chap1.pdf, p.7)*

A1.4 At $f = 1$ GHz in air: $\lambda = v_0/f = 0.3$ m. $k = 0.05/0.3 = 0.167$. Since $k > 0.1$ and approaching 0.2, transmission-line analysis is advisable. *(chap1.pdf, p.21)*

A1.5 Two structures with the same physical size but in different media have different electrical sizes ($k = \mathscr{L}/\lambda$, and λ changes with ϵ_r). A structure's ability to radiate and its need for distributed analysis depends on k, not physical length. *(chap1.pdf, p.15, 21)*

A1.6 (a) $v = v_0/\sqrt{4} = 1.5 \times 10^8$ m/s. (b) $T_D = 0.20/(1.5 \times 10^8) = 1.33$ ns. (c) $\lambda = v/f = 1.5 \times 10^8/(7.5 \times 10^8) = 0.2$ m. (d) $k = 0.20/0.20 = 1.0$. The trace is one full wavelength long — strongly distributed. *(chap1.pdf, p.16–21)*

A1.7 (a) $20\log_{10}(100) = 40$ dB. (b) $10\log_{10}(0.001) = -30$ dB. (c) $-14+107 = 93$ dBμV. (d) $87 - 107 = -20$ dBm. *(chap1.pdf, p.25–27, 30)*

A1.8 Cable loss $= 8.686 \times 0.08 \times 5 = 3.47$ dB. *(chap1.pdf, p.30)*

A1.9 Total gain $= 20 - 6 + 14 = 28$ dB. Output $= -10 + 28 = 18$ dBm. *(chap1.pdf, p.28)*

A1.10 False. In a denser dielectric, v is smaller, so $\lambda = v/f$ is **shorter**. *(chap1.pdf, p.20–21)*

A1.11 False. Doubling the ratio adds 6 dB ($20\log_{10} 2 \approx 6$), not doubles the dB value. *(chap1.pdf, p.26–27)*

A1.12 False. $T_D = \mathscr{L}/v$ depends on length and medium, not frequency (for lossless line). *(chap1.pdf, p.16)*

A1.13 True. *(chap1.pdf, p.6)*

A1.14 (C) 1.0. At 3 GHz, $\lambda = 0.1$ m in air. $k = 0.10/0.10 = 1.0$. *(chap1.pdf, p.21)*

A1.15 (D). When $R_{\text{in}} = R_L$, power gain in dB $= 20\log_{10}(V_{\text{out}}/V_{\text{in}})$, same as voltage gain. *(chap1.pdf, p.26)*

Chapter 2

A2.1 Class A = commercial/industrial; Class B = residential. Class B is **more stringent** because residential users lack technical expertise or resources to resolve interference problems. *(chap2.pdf, p.6)*

A2.2 (1) Conducted emissions: noise via AC power cord, 150 kHz–30 MHz, measured with LISN. (2) Radiated emissions: electromagnetic radiation, 30 MHz–40 GHz, measured in SAC or OATS. *(chap2.pdf, p.7)*

A2.3 The LISN is inserted between the power net and EUT. It (1) blocks 60 Hz from reaching the receiver, and (2) passes 150 kHz–30 MHz noise to the receiver. Its ~50 Ω impedance ensures standardized, repeatable measurements. *(chap2.pdf, p.29, 32)*

A2.4 FCC radiated: 30 MHz–40 GHz, Class B at 3 m, Class A at 10 m. CISPR 22 radiated: 30 MHz–1 GHz, both classes at 10 m. *(chap2.pdf, p.11, 14)*

A2.5 FCC and CISPR 22 are legal requirements that cannot be waived — non-compliance means a product cannot be sold. MIL-STD-461E can be tailored or waived by the procuring military authority for specific platforms or applications. *(chap2.pdf, p.5, 17)*

A2.6 $\Delta E = 20\log_{10}(10/3) \approx 10.46$ dB. Class A at 3 m: $39 + 10.46 = 49.46$ dBμV/m. Class B at 3 m: 40 dBμV/m. Class B is **more stringent** (lower limit = stricter). *(chap2.pdf, p.13)*

A2.7 The margin at 3 m is **not the same** as at 10 m — FCC Class B is measured at 3 m, Class A at 10 m. A Class A product passing at 10 m by 5 dB means it is 5 dB below 39 dBμV/m = 34 dBμV/m at 10 m = 44.5 dBμV/m at 3 m. The Class B limit at 3 m is 40 dBμV/m. The product **fails** Class B by 4.5 dB. *(chap2.pdf, p.11, 13)*

A2.8 Measured: 38.5 dBμV/m; limit: 43.5 dBμV/m. Margin $= 43.5 - 38.5 = 5$ dB below limit → **passes by 5 dB**. *(chap2.pdf, p.11, 33)*

A2.9 True. *(chap2.pdf, p.15)*

A2.10 True. When normalized to 3 m, Class A = Class A (10 m) + 10.5 dB, which is still about 10 dB above the Class B limit. *(chap2.pdf, p.13)*

A2.11 False. CISPR 22 covers up to **1 GHz**; FCC goes up to 40 GHz. *(chap2.pdf, p.14)*

A2.12 True (for aircraft external/safety-critical: 200 V/m). *(chap2.pdf, p.20)*

A2.13 (B) 150 kHz – 30 MHz. *(chap2.pdf, p.7)*

A2.14 (C) +10.5 dB ($= 20\log_{10}(10/3)$). *(chap2.pdf, p.13)*

Chapter 3

A3.1 The spectral envelope has two breakpoints: $f_1 = 1/(\pi\tau)$ from the pulse width, and $f_2 = 1/(\pi\tau_r)$ from the rise time. Above f_2, the envelope falls at -40 dB/decade. The clock frequency determines the harmonic spacing ($f_0 = 1/T$) but does not change f_2. Faster edges (smaller τ_r) push f_2 higher, generating more energy at high frequencies. *(chap3.pdf, p.3, 34)*

A3.2 Sketch: flat at $2A\tau/T$ for $f < f_1$; drops at -20 dB/dec between f_1 and f_2; drops at -40 dB/dec above f_2. Label $f_1 = 1/(\pi\tau)$ and $f_2 = 1/(\pi\tau_r)$. *(chap3.pdf, p.34)*

A3.3 For 50% duty cycle, $\tau = T/2$. The first sinc factor in c_n becomes $|\sin(n\pi\tau/T)| = |\sin(n\pi/2)|$, which is zero for all even n. *(chap3.pdf, p.14)*

A3.4 Reducing duty cycle moves $f_1 = 1/(\pi\tau)$ higher (since τ is smaller), but $f_2 = 1/(\pi\tau_r)$ depends only on τ_r, which is unchanged. The high-frequency roll-off (above f_2) is identical regardless of duty cycle. *(chap3.pdf, p.39)*

A3.5 A peak detector captures the peak amplitude (slow discharge). A quasi-peak detector has a discharge resistor so its reading falls between peak and average — it weights pulsed interference less than CW. QP is used in regulations because it better represents the subjective nuisance of different interference types. *(chap3.pdf, p.46)*

A3.6 The PSD has: (1) a delta function at DC representing the average power ($X_0^2/4$ per unit bandwidth); (2) a continuous sinc-squared component representing the power distributed across frequency due to random bit transitions. The DC delta function exists because even random data has a nonzero average value ($X_0/2$). *(chap3.pdf, p.52)*

A3.7 (a) $f_0 = 1/T = 50$ MHz; $f_1 = 1/(\pi \times 8 \times 10^{-9}) = 39.8$ MHz; $f_2 = 1/(\pi \times 2 \times 10^{-9}) = 159$ MHz; BW $= 1/\tau_r = 500$ MHz. (b) $c_0 = A\tau/T = 5 \times 8/20 = 2$ V. (c) -40 dB/decade. *(chap3.pdf, p.34, 37)*

A3.8 $|c_n| = \frac{A\tau}{T}\left|\frac{\sin(\pi n\tau/T)}{n\pi\tau/T}\right|$ where $A = 3$ V, $\tau = 10$ ns, $T = 100$ ns.

(a) $n = 1$: $|c_1| = \frac{3\times 10}{100}\left|\frac{\sin(\pi\times 0.1)}{\pi\times 0.1}\right| = 0.3\times\left|\frac{\sin(0.314)}{0.314}\right| = 0.3\times 0.984 = 0.295$ V.

(b) $n = 5$: $|c_5| = 0.3\times\left|\frac{\sin(5\pi\times 0.1)}{5\pi\times 0.1}\right| = 0.3\times\left|\frac{\sin(\pi/2)}{\pi/2}\right| = 0.3\times\frac{2}{\pi} = 0.191$ V.

(chap3.pdf, p.12)

A3.9 (a) Bit rate $= 1/T = 1/10^{-9} = 1$ Gbps. (b) First null at $f = 1/T = 1$ GHz. (c) $G_x(f) = \frac{(3.3)^2}{4}\delta(f) + \frac{(3.3)^2\times 10^{-9}}{4}\frac{\sin^2(\pi f\times 10^{-9})}{(\pi f\times 10^{-9})^2} = 2.72\,\delta(f) + 2.72\times 10^{-9}\frac{\sin^2(\pi f\times 10^{-9})}{(\pi f\times 10^{-9})^2}$ W/Hz. *(chap3.pdf, p.52)*

A3.10 True. Higher $\tau_r \to$ lower $f_2 = 1/(\pi\tau_r) \to$ the -40 dB/decade slope starts sooner $\to$ less energy at high frequencies. *(chap3.pdf, p.35)*

A3.11 True. *(chap3.pdf, p.52)*

A3.12 False. BW $= 1/\tau_r$ (rise time, not clock period). *(chap3.pdf, p.37)*

A3.13 False. For 50% duty cycle, **even harmonics** are zero and odd harmonics are nonzero. *(chap3.pdf, p.14)*

A3.14 (C) 2 GHz. BW $= 1/\tau_r = 1/(0.5\times 10^{-9}) = 2\times 10^9$ Hz. *(chap3.pdf, p.37)*

A3.15 (C) -40 dB/decade. *(chap3.pdf, p.34)*

A3.16 (C) stays the same. $f_2 = 1/(\pi\tau_r)$ depends only on τ_r, not τ. *(chap3.pdf, p.34)*

Chapter 4

A4.1 KVL: $\partial V/\partial z = -l\,\partial I/\partial t$ *(chap4.pdf, p.8)*; KCL: $\partial I/\partial z = -c\,\partial V/\partial t$ *(chap4.pdf, p.9)*. KVL gives the voltage equation (series inductance drops voltage); KCL gives the current equation (shunt capacitance draws current).

A4.2 TEM = Transverse ElectroMagnetic. There are no electric or magnetic field components in the direction of propagation ($E_z = 0$, $H_z = 0$). This allows the static-field formulas for l and c to be used. *(chap4.pdf, p.11)*

A4.3 $Z_C = \sqrt{l/c}$ is the ratio of voltage to current for a single traveling wave. For signal integrity, if the load impedance matches Z_C, the wave is fully absorbed — no reflections, no distortion. Any mismatch causes partial reflection, creating signal distortion and overshoot. *(chap4.pdf, p.28, 34)*

A4.4 Open ($R_L \to \infty$): $\Gamma_L = +1$, wave reflects with same polarity, voltage doubles at load. Short ($R_L = 0$): $\Gamma_L = -1$, wave reflects with reversed polarity, voltage is zero at load. Matched ($R_L = Z_C$): $\Gamma_L = 0$, no reflection, clean waveform at load. *(chap4.pdf, p.35)*

A4.5 The line can be ignored when $\tau_r > 10T_D$. If rise time is at least 10 times the one-way travel time, the line appears as a lumped element. *(chap4.pdf, p.59)*

A4.6 Series match ($R_S + R = Z_C$): eliminates reflections; no power dissipated in R at steady state; load voltage may reach full V_0. Parallel match ($R \| R_L = Z_C$): eliminates reflections; continuously dissipates power; load voltage is reduced. *(chap4.pdf, p.54–55)*

A4.7 Current at DC is uniformly distributed across the conductor cross-section. As f increases, skin effect concentrates current in a thin annular shell of thickness $\delta = 1/\sqrt{\pi f \mu_0 \sigma}$. Since $\delta \propto 1/\sqrt{f}$, the effective conducting area decreases, so resistance increases as $\sqrt{f}$ (10 dB/decade). *(chap4.pdf, p.82–83)*

A4.8 Loss tangent: $\tan\theta = \epsilon_r''/\epsilon_r'$. It gives the dielectric conductance $g = \omega c \tan\theta$, which increases **linearly with** f (20 dB/decade). Skin-effect resistance r increases as $\sqrt{f}$ (10 dB/decade). So at high enough frequency, dielectric loss dominates over conductor loss. *(chap4.pdf, p.87)*

A4.9 (a) $\Gamma_L = (25-75)/(25+75) = -50/100 = -0.5$. $\Gamma_S = (150-75)/(150+75) = 75/225 = 1/3$. (b) $V^+ = \frac{75}{150+75} \times 6 = \frac{75}{225} \times 6 = 2$ V. (c) $V(\mathscr{L}, T_D^+) = V^+(1+\Gamma_L) = 2 \times (1-0.5) = 1$ V. *(chap4.pdf, p.35, 37)*

A4.10 (a) $\beta = \omega/v = 2\pi \times 10^9/(2 \times 10^8) = 10\pi = 31.4$ rad/m. (b) $\beta\mathscr{L} = 31.4 \times 0.5 = 15.7$ rad $= 5\pi$ rad. (c) Open circuit: $Z_L \to \infty$. $Z_{\text{in}} = Z_C/(j \tan\beta\mathscr{L}) = 50/(j \tan 5\pi) = 50/(j \times 0)$. Since $\tan(5\pi) = 0$: $Z_{\text{in}} \to \infty$ — open circuit (makes sense for line that is 2.5 wavelengths long). *(chap4.pdf, p.74, 76)*

A4.11 $\Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C} = \frac{(100+j50)-50}{(100+j50)+50} = \frac{50+j50}{150+j50}$.

$|\Gamma_L| = \frac{|50+j50|}{|150+j50|} = \frac{\sqrt{50^2+50^2}}{\sqrt{150^2+50^2}} = \frac{70.7}{158.1} = 0.447$.

VSWR $= (1+0.447)/(1-0.447) = 1.447/0.553 = 2.62$. *(chap4.pdf, p.74, 77)*

A4.12 $T_D = \mathscr{L}/v = 0.10/(2 \times 10^8) = 0.5$ ns.

(a) $\tau_r = 5$ ns, $10T_D = 5$ ns. $\tau_r = 10T_D$ — on the boundary; line effects are just beginning to matter.

(b) $\tau_r = 0.5$ ns, $10T_D = 5$ ns. $\tau_r < 10T_D$ — **line matters significantly**.

(chap4.pdf, p.59)

A4.13 Series match requires $R_S + R = Z_C$, so $R = Z_C - R_S = 50 - 20 = 30\,\Omega$. *(chap4.pdf, p.54)*

A4.14 $\delta = 1/\sqrt{\pi f \mu_0 \sigma}$ where $\mu_0 = 4\pi \times 10^{-7}$ H/m, $\sigma = 5.8 \times 10^7$ S/m.

(a) At 1 MHz: $\delta = 1/\sqrt{\pi \times 10^6 \times 4\pi \times 10^{-7} \times 5.8 \times 10^7} = 1/\sqrt{4\pi^2 \times 5.8 \times 10^6} = 1/\sqrt{2.285 \times 10^8} = 66\,\mu$m.

(b) At 1 GHz: $\delta = 66\,\mu\text{m}/\sqrt{1000} = 66/31.6 \approx 2.1\,\mu$m.

Skin depth decreases by $\sqrt{1000} \approx 31.6\times$ when frequency increases $1000\times$. *(chap4.pdf, p.82)*

A4.15 $\alpha \approx \frac{1}{2}(r/Z_C + gZ_C) = \frac{1}{2}(0.5/50 + 10^{-4} \times 50) = \frac{1}{2}(0.01 + 0.005) = 0.0075$ Np/m.

Cable loss $= 8.686 \times 0.0075 \times 3 = 0.196$ dB. *(chap4.pdf, p.91, 89)*

A4.16 True. *(chap4.pdf, p.35)*

A4.17 False. $lc = \mu\epsilon$ — the product depends entirely on the surrounding medium (μ and ϵ), not the geometry. *(chap4.pdf, p.12)*

A4.18 True. *(chap4.pdf, p.43)*

A4.19 True. *(chap4.pdf, p.77)*

A4.20 False. Skin depth **decreases** as $1/\sqrt{f}$ (i.e., $\delta \propto f^{-1/2}$). *(chap4.pdf, p.82)*

A4.21 True. *(chap4.pdf, p.91)*

A4.22 (B) $\Gamma_L = -1$, VSWR $= \infty$. *(chap4.pdf, p.35, 77)*

A4.23 (B) KVL around a loop. *(chap4.pdf, p.8)*

A4.24 (B) Can be treated as lumped — $\tau_r = 2$ ns, $10T_D = 1$ ns, so $\tau_r > 10T_D$. *(chap4.pdf, p.59)*

A4.25 (C) $\propto \sqrt{f}$ (10 dB/decade). *(chap4.pdf, p.83)*